Supplementary Material for "Beyond Local Centers toward Prototypes: Joint Learning of Self-Representation and Adaptive Clustering"

Dahyun Jeong
University of Seoul
Seoul, Republic of Korea
dahyunjeong@uos.ac.kr

Yunjin Choi
University of Seoul
Seoul, Republic of Korea
ycstat@uos.ac.kr

1 Implementation

1.1 Multi-Affine ADMM

Let $\mathbf{X} \in \mathbb{R}^{n\times p}$ denote the observed data matrix, where the i-th row $\mathbf{x}_i \in \mathbb{R}^p$ represents the i-th observation. We introduce a prototype matrix $\mathbf{U} \in \mathbb{R}^{n\times p}$, whose i-th row $\mathbf{u}_i$ is the learned representative associated with $\mathbf{x}_i$. We also introduce an adaptive proximity matrix $\mathbf{Z} \in \mathbb{R}^{n\times n}$, where $\mathbf{Z}_{ij}$ represents the learned connection strength from observation i to observation j. Our optimization problem is:

$$\begin{aligned}\min_{\mathbf{U},\,\mathbf{Z}} \quad & \frac{1}{2}\|\mathbf{X}-\mathbf{U}\|_F^2 + \gamma\|\mathbf{Z}\|_F^2 + \lambda\sum_{i<j}\left(\mathbf{Z}_{ij}\|\mathbf{U}_{i\cdot}-\mathbf{U}_{j\cdot}\|_2 + \mathbf{Z}_{ij}^2\|\mathbf{U}_{i\cdot}-\mathbf{U}_{j\cdot}\|_2^2\right)\\ \text{such that} \quad & \mathbf{Z}\mathbf{U}=\mathbf{U}, \quad \operatorname{diag}(\mathbf{Z})=0, \quad \mathbf{Z}\geq\mathbf{0}, \quad \mathbf{Z}=\mathbf{Z}^\top, \quad \mathbf{Z}\mathbf{1}=\mathbf{1}.\end{aligned}$$

Here, $\lambda,\ \gamma>0$, $\mathbf{U}_{i\cdot}$ denotes the i-th row of $\mathbf{U}$, and $\mathbf{Z}_{ij}$ denotes the (i,j)-th element of $\mathbf{Z}$.

The proposed optimization problem is nonconvex due to its constraints. We therefore solve the proposed optimization problem using a multi-affine ADMM framework (Gao et al., 2020). Specifically, the proposed optimization can be reformulated as

$$\begin{aligned}\min_{\substack{\mathbf{U},\mathbf{U}_1,\mathbf{U}_2,\mathbf{U}_3,\\ \mathbf{Z},\mathbf{Z}_1,\mathbf{Z}_2,\mathbf{Z}_3,\mathbf{Z}_4,\mathbf{A}}} \quad & \frac{1}{2}\|\mathbf{X}-\mathbf{U}\|_F^2 + \gamma\|\mathbf{Z}\|_F^2 + \lambda\sum_{\ell\in\mathcal{E}}\left(\|\mathbf{A}_{\ell\cdot}\|_2 + \|\mathbf{A}_{\ell\cdot}\|_F^2\right)\\ & \quad + I_{\mathcal{C}_1}(\mathbf{Z}_1) + I_{\mathcal{C}_2}(\mathbf{Z}_2)\\ \text{s.t.} \quad & \mathbf{Z}_1=\mathbf{Z}_2, \quad \mathbf{Z}_3=\mathbf{Z}, \quad \mathbf{Z}_4=\mathbf{Z}, \quad \frac{1}{2}(\mathbf{Z}_1+\mathbf{Z}_2^\top)=\mathbf{Z},\\ & \mathbf{U}_1=\mathbf{U}, \quad \mathbf{U}_2=\mathbf{U}, \quad \mathbf{U}_3=\mathbf{U},\\ & \mathbf{Z}_3\mathbf{U}_1=\mathbf{U},\\ & (\mathbf{Z}_4)_{\ell_1\ell_2}\left\{(\mathbf{U}_2)_{\ell_1\cdot}-(\mathbf{U}_3)_{\ell_2\cdot}\right\}=\mathbf{A}_{\ell\cdot} \quad \text{for } \ell\in\mathcal{E}\end{aligned}$$

where

$$\begin{aligned}\mathcal{C}_1 &= \left\{\mathbf{M}\in\mathbb{R}_+^{n\times n} : \operatorname{diag}(\mathbf{M})=0,\ \mathbf{M}\mathbf{1}=\mathbf{1}\right\},\\ \mathcal{C}_2 &= \left\{\mathbf{M}\in\mathbb{R}_+^{n\times n} : \operatorname{diag}(\mathbf{M})=0,\ \mathbf{M}^\top\mathbf{1}=\mathbf{1}\right\}\end{aligned}$$

encodes the constraints on the self-representation matrix $\mathbf{Z}$, and

$$\mathcal{E} = \{(\ell_1,\ell_2) : \ell=1,\cdots,L,\ \ell_1<\ell_2\}.$$

denote the set of all pairwise sample indices. Here, $\mathbf{A}_{l\cdot}$ denotes the lth row of a $|\mathcal{E}|\times p$ matrix $\mathbf{A}$ which reparametrizes the weighted pairwise difference $(\mathbf{Z}_4)_{\ell_1\ell_2}\left\{(\mathbf{U}_2)_{\ell_1\cdot}-(\mathbf{U}_3)_{\ell_2\cdot}\right\}$.

To satisfy the regularity conditions required by multi-affine ADMM, we introduce slack variables $\mathbf{E}_1, \cdots, \mathbf{E}_9$ and add quadratic regularization terms for the slack variables. The resulting regularized problem is

$$
\begin{aligned}
\min_{\substack{\mathbf{U},\mathbf{U}_1,\mathbf{U}_2,\mathbf{U}_3,\\ \mathbf{Z},\mathbf{Z}_1,\mathbf{Z}_2,\mathbf{Z}_3,\mathbf{Z}_4,\mathbf{A},\\ \mathbf{E}_1,\cdots,\mathbf{E}_9}} \quad & \frac{1}{2}\|\mathbf{X}-\mathbf{U}\|_F^2 + \gamma\|\mathbf{Z}\|_F^2 + \lambda\sum_{\ell=1}^{L}\left(\|\mathbf{A}_{\ell\cdot}\|_2 + \frac{1}{2}\|\mathbf{A}_{\ell\cdot}\|_2^2\right) \\
& + I_{\mathcal{C}_1}(\mathbf{Z}_1) + I_{\mathcal{C}_2}(\mathbf{Z}_2) + \frac{\mu}{2}\sum_{k=1}^{9}\|\mathbf{E}_k\|_F^2 \\
\text{s.t.} \quad & \mathbf{Z}_1 - \mathbf{Z}_2 = \mathbf{E}_1, \quad \mathbf{Z}_3 - \mathbf{Z} = \mathbf{E}_2, \quad \mathbf{Z}_4 - \mathbf{Z} = \mathbf{E}_3 \\
& \frac{1}{2}(\mathbf{Z}_1 + \mathbf{Z}_2^\top) - \mathbf{Z} = \mathbf{E}_4, \\
& \mathbf{U}_1 - \mathbf{U} = \mathbf{E}_5, \quad \mathbf{U}_2 - \mathbf{U} = \mathbf{E}_6, \quad \mathbf{U}_3 - \mathbf{U} = \mathbf{E}_7, \\
& \mathbf{Z}_3\mathbf{U}_1 - \mathbf{U} = \mathbf{E}_8, \\
& (\mathbf{Z}_4)_{\ell_1\ell_2}\left\{(\mathbf{U}_2)_{\ell_1\cdot} - (\mathbf{U}_3)_{\ell_2\cdot}\right\} - \mathbf{A}_{\ell\cdot} = (\mathbf{E}_9)_{\ell\cdot}, \text{ for } \ell \in \mathcal{E}.
\end{aligned}
$$

The parameter $\mu > 0$ determines the strength of the quadratic penalty imposed on the slack variables. As μ increases, deviations of the slack variables from zero are more strongly penalized, thereby making the regularized slack formulation a closer approximation to the original constrained problem. In our analysis, we increase μ until the norm of the slack variables falls below numerical precision.

The scaled augmented Lagrangian is given by

$$
\begin{aligned}
\mathcal{L}_\rho =& \frac{1}{2}\|\mathbf{X}-\mathbf{U}\|_F^2 + \gamma\|\mathbf{Z}\|_F^2 + I_{\mathcal{C}_1}(\mathbf{Z}_1) + I_{\mathcal{C}_2}(\mathbf{Z}_2) \\
& + \lambda\sum_{\ell=1}^{L}\left(\|\mathbf{A}_{\ell\cdot}\|_2 + \frac{1}{2}\|\mathbf{A}_{\ell\cdot}\|_2^2\right) + \frac{\mu}{2}\sum_{k=1}^{9}\|\mathbf{E}_k\|_F^2 \\
& + \frac{\rho}{2}\Bigg[\left\|\frac{1}{2}(\mathbf{Z}_1 + \mathbf{Z}_2^\top) - \mathbf{Z} - \mathbf{E}_1 + \mathbf{\Psi}_1\right\|_F^2 + \|\mathbf{Z}_1 - \mathbf{Z}_2 - \mathbf{E}_2 + \mathbf{\Psi}_2\|_F^2 \\
& \quad + \|\mathbf{Z}_3\mathbf{U}_1 - \mathbf{U} - \mathbf{E}_3 + \mathbf{\Psi}_3\|_F^2 + \|\mathbf{Z}_3 - \mathbf{Z} - \mathbf{E}_4 + \mathbf{\Psi}_4\|_F^2 + \|\mathbf{U}_1 - \mathbf{U} - \mathbf{E}_5 + \mathbf{\Psi}_5\|_F^2 \\
& \quad + \sum_{\ell=1}^{L}\|(\mathbf{Z}_4)_{\ell_1\ell_2}\left\{(\mathbf{U}_2)_{\ell_1\cdot} - (\mathbf{U}_3)_{\ell_2\cdot}\right\} - \mathbf{A}_{\ell\cdot} - (\mathbf{E}_6)_{\ell\cdot} + (\mathbf{\Psi}_6)_{\ell\cdot}\|_2^2 \\
& \quad + \|\mathbf{U}_2 - \mathbf{U} - \mathbf{E}_7 + \mathbf{\Psi}_7\|_F^2 + \|\mathbf{U}_3 - \mathbf{U} - \mathbf{E}_8 + \mathbf{\Psi}_8\|_F^2 + \|\mathbf{Z}_4 - \mathbf{Z} - \mathbf{E}_9 + \mathbf{\Psi}_9\|_F^2\Bigg],
\end{aligned}
$$

where $\rho > 0$ is the ADMM penalty parameter and $\{\mathbf{\Psi}_k\}_{k=1}^9$ are scaled dual variables. Since the formulation contains several consensus and coupling constraints, we use a relatively large value of ρ in practice to encourage feasibility during the ADMM iterations.

1.2 Update Rule

Update Z The $\mathbf{Z}$-subproblem is

$$
\begin{aligned}
\min_{\mathbf{Z}} \quad & \gamma\|\mathbf{Z}\|_F^2 + \frac{\rho}{2}\left\|\frac{1}{2}(\mathbf{Z}_1 + \mathbf{Z}_2^\top) - \mathbf{Z} - \mathbf{E}_1 + \mathbf{\Psi}_1\right\|_F^2 \\
& + \frac{\rho}{2}\|\mathbf{Z}_3 - \mathbf{Z} - \mathbf{E}_4 + \mathbf{\Psi}_4\|_F^2 + \frac{\rho}{2}\|\mathbf{Z}_4 - \mathbf{Z} - \mathbf{E}_9 + \mathbf{\Psi}_9\|_F^2.
\end{aligned}
$$

Hence,

$$
\mathbf{Z} = \frac{\rho}{2\gamma + 3\rho}\left[\frac{1}{2}(\mathbf{Z}_1 + \mathbf{Z}_2^\top) + \mathbf{Z}_3 + \mathbf{Z}_4 - \mathbf{E}_1 - \mathbf{E}_4 - \mathbf{E}_9 + \mathbf{\Psi}_1 + \mathbf{\Psi}_4 + \mathbf{\Psi}_9\right]. \tag{1}
$$

Update $\mathbf{Z}_1$ and $\mathbf{Z}_2$ The $\mathbf{Z}_1$-subproblem is

$$\begin{aligned}\min_{\mathbf{Z}_1} \quad & \frac{1}{2}\left\|\frac{1}{2}(\mathbf{Z}_1+\mathbf{Z}_2^\top)-\mathbf{Z}-\mathbf{E}_1+\mathbf{\Psi}_1\right\|_F^2+\frac{1}{2}\|\mathbf{Z}_1-\mathbf{Z}_2-\mathbf{E}_2+\mathbf{\Psi}_2\|_F^2 \\ \text{s.t.} \quad & \operatorname{diag}(\mathbf{Z}_1)=0, \qquad \mathbf{Z}_1\geq 0, \qquad \mathbf{Z}_1\mathbf{1}=\mathbf{1}.\end{aligned}$$

The unconstrained minimizer is

$$\frac{4}{5}\left(\frac{1}{2}\mathbf{Z}+\mathbf{Z}_2-\frac{1}{4}\mathbf{Z}_2^\top+\frac{1}{2}\mathbf{E}_1+\mathbf{E}_2-\frac{1}{2}\mathbf{\Psi}_1-\mathbf{\Psi}_2\right).$$

Since the feasible set is row-separable with the diagonal fixed to zero, the update is computed row by row. For $i=1,\cdots,n$, define

$$\mathbf{y}_i^{(1)}:=\left[\frac{4}{5}\left(\frac{1}{2}Z_{ij}+(\mathbf{Z}_2)_{ij}-\frac{1}{4}(\mathbf{Z}_2)_{ji}+\frac{1}{2}(\mathbf{E}_1)_{ij}+(\mathbf{E}_2)_{ij}-\frac{1}{2}(\mathbf{\Psi}_1)_{ij}-(\mathbf{\Psi}_2)_{ij}\right)\right]_{j\neq i}.$$

Then

$$(\mathbf{Z}_1)_{ii}=0, \qquad [(\mathbf{Z}_1)_{ij}]_{j\neq i}=\textsc{ProjectToSimplex}\left(\mathbf{y}_i^{(1)}\right). \tag{2}$$

Similarly, the $\mathbf{Z}_2$-update is obtained by projection onto the column-wise simplex constraint:

$$(\mathbf{Z}_2)_{jj}=0, \qquad [(\mathbf{Z}_2)_{ij}]_{i\neq j}=\textsc{ProjectToSimplex}\left(\mathbf{y}_j^{(2)}\right), \tag{3}$$

where $\mathbf{y}_j^{(2)}$ is defined analogously from the unconstrained minimizer of the $\mathbf{Z}_2$-subproblem.

Update $\mathbf{Z}_3$ The $\mathbf{Z}_3$-subproblem is

$$\min_{\mathbf{Z}_3} \quad \frac{1}{2}\|\mathbf{Z}_3\mathbf{U}_1-\mathbf{U}-\mathbf{E}_3+\mathbf{\Psi}_3\|_F^2+\frac{1}{2}\|\mathbf{Z}_3-\mathbf{Z}-\mathbf{E}_4+\mathbf{\Psi}_4\|_F^2.$$

Therefore,

$$\mathbf{Z}_3=\left[(\mathbf{U}+\mathbf{E}_3-\mathbf{\Psi}_3)\mathbf{U}_1^\top+\mathbf{Z}+\mathbf{E}_4-\mathbf{\Psi}_4\right](\mathbf{U}_1\mathbf{U}_1^\top+\mathbf{I}_n)^{-1}. \tag{4}$$

Update $(\mathbf{Z}_4)_{\ell_1\ell_2}$ for each edge ℓ For each edge $\ell=(\ell_1,\ell_2)$, the subproblem is

$$\begin{aligned}\min_{(\mathbf{Z}_4)_{\ell_1\ell_2}} \quad & \frac{1}{2}\left\|(\mathbf{Z}_4)_{\ell_1\ell_2}\{(\mathbf{U}_2)_{\ell_1\cdot}-(\mathbf{U}_3)_{\ell_2\cdot}\}-\mathbf{A}_{\ell\cdot}-(\mathbf{E}_6)_{\ell\cdot}+(\mathbf{\Psi}_6)_{\ell\cdot}\right\|_2^2 \\ & +\frac{1}{2}\left((\mathbf{Z}_4)_{\ell_1\ell_2}-Z_{\ell_1\ell_2}-(\mathbf{E}_9)_{\ell_1\ell_2}+(\mathbf{\Psi}_9)_{\ell_1\ell_2}\right)^2.\end{aligned}$$

Thus,

$$(\mathbf{Z}_4)_{\ell_1\ell_2}=\frac{\{\mathbf{A}_{\ell\cdot}+(\mathbf{E}_6)_{\ell\cdot}-(\mathbf{\Psi}_6)_{\ell\cdot}\}\{(\mathbf{U}_2)_{\ell_1\cdot}-(\mathbf{U}_3)_{\ell_2\cdot}\}^\top+\mathbf{Z}_{\ell_1\ell_2}+(\mathbf{E}_9)_{\ell_1\ell_2}-(\mathbf{\Psi}_9)_{\ell_1\ell_2}}{\|(\mathbf{U}_2)_{\ell_1\cdot}-(\mathbf{U}_3)_{\ell_2\cdot}\|_2^2+1}. \tag{5}$$

Update $\mathbf{U}$ The $\mathbf{U}$-subproblem is

$$\begin{aligned}\min_{\mathbf{U}} \quad & \frac{1}{2}\|\mathbf{X}-\mathbf{U}\|_F^2+\frac{\rho}{2}\|\mathbf{Z}_3\mathbf{U}_1-\mathbf{U}-\mathbf{E}_3+\mathbf{\Psi}_3\|_F^2+\frac{\rho}{2}\|\mathbf{U}_1-\mathbf{U}-\mathbf{E}_5+\mathbf{\Psi}_5\|_F^2 \\ & +\frac{\rho}{2}\|\mathbf{U}_2-\mathbf{U}-\mathbf{E}_7+\mathbf{\Psi}_7\|_F^2+\frac{\rho}{2}\|\mathbf{U}_3-\mathbf{U}-\mathbf{E}_8+\mathbf{\Psi}_8\|_F^2.\end{aligned}$$

Hence,

$$\begin{aligned}\mathbf{U}=\frac{1}{1+4\rho}\Big[\mathbf{X}+\rho\big(&\mathbf{Z}_3\mathbf{U}_1+\mathbf{U}_1+\mathbf{U}_2+\mathbf{U}_3 \\ &-\mathbf{E}_3-\mathbf{E}_5-\mathbf{E}_7-\mathbf{E}_8+\mathbf{\Psi}_3+\mathbf{\Psi}_5+\mathbf{\Psi}_7+\mathbf{\Psi}_8\big)\Big].\end{aligned} \tag{6}$$

Update $\mathbf{U}_1$ The $\mathbf{U}_1$-subproblem is

$$\min_{\mathbf{U}_1} \quad \frac{1}{2}\|\mathbf{Z}_3\mathbf{U}_1-\mathbf{U}-\mathbf{E}_3+\mathbf{\Psi}_3\|_F^2+\frac{1}{2}\|\mathbf{U}_1-\mathbf{U}-\mathbf{E}_5+\mathbf{\Psi}_5\|_F^2.$$

Therefore,

$$\mathbf{U}_1=(\mathbf{Z}_3^\top\mathbf{Z}_3+\mathbf{I}_n)^{-1}\left[\mathbf{Z}_3^\top(\mathbf{U}+\mathbf{E}_3-\mathbf{\Psi}_3)+\mathbf{U}+\mathbf{E}_5-\mathbf{\Psi}_5\right]. \tag{7}$$

Update $(\mathbf{U}_2)_{i\cdot}$ and $(\mathbf{U}_3)_{j\cdot}$ Let

$$\mathcal{N}_1(i) := \{\ell \in \{1, \cdots, L\} : \ell_1 = i\}, \qquad \mathcal{N}_2(j) := \{\ell \in \{1, \cdots, L\} : \ell_2 = j\}.$$

For fixed i and j, the subproblems are solved row-wise. The closed-form updates are

$$(\mathbf{U}_2)_{i\cdot} = \frac{\mathbf{U}_{i\cdot} + (\mathbf{E}_7)_{i\cdot} - (\mathbf{\Psi}_7)_{i\cdot} + \sum_{\ell \in \mathcal{N}_1(i)} \left[(\mathbf{Z}_4)^2_{\ell_1\ell_2}(\mathbf{U}_3)_{\ell_2\cdot} + (\mathbf{Z}_4)_{\ell_1\ell_2}\left\{\mathbf{A}_{\ell\cdot} + (\mathbf{E}_6)_{\ell\cdot} - (\mathbf{\Psi}_6)_{\ell\cdot}\right\}\right]}{1 + \sum_{\ell \in \mathcal{N}_1(i)}(\mathbf{Z}_4)^2_{\ell_1\ell_2}}, \tag{8}$$

and

$$(\mathbf{U}_3)_{j\cdot} = \frac{\mathbf{U}_{j\cdot} + (\mathbf{E}_8)_{j\cdot} - (\mathbf{\Psi}_8)_{j\cdot} + \sum_{\ell \in \mathcal{N}_2(j)} \left[(\mathbf{Z}_4)^2_{\ell_1\ell_2}(\mathbf{U}_2)_{\ell_1\cdot} - (\mathbf{Z}_4)_{\ell_1\ell_2}\left\{\mathbf{A}_{\ell\cdot} + (\mathbf{E}_6)_{\ell\cdot} - (\mathbf{\Psi}_6)_{\ell\cdot}\right\}\right]}{1 + \sum_{\ell \in \mathcal{N}_2(j)}(\mathbf{Z}_4)^2_{\ell_1\ell_2}}. \tag{9}$$

Update $\mathbf{A}_{\ell\cdot}$ for each edge ℓ For a fixed edge ℓ, the subproblem is

$$\begin{aligned}\min_{\mathbf{A}_{\ell\cdot}} \quad & \lambda\|\mathbf{A}_{\ell\cdot}\|_2 + \frac{\lambda}{2}\|\mathbf{A}_{\ell\cdot}\|_2^2 \\ & + \frac{\rho}{2}\left\|(\mathbf{Z}_4)_{\ell_1\ell_2}\left\{(\mathbf{U}_2)_{\ell_1\cdot} - (\mathbf{U}_3)_{\ell_2\cdot}\right\} - \mathbf{A}_{\ell\cdot} - (\mathbf{E}_6)_{\ell\cdot} + (\mathbf{\Psi}_6)_{\ell\cdot}\right\|_2^2.\end{aligned}$$

Therefore, $\mathbf{A}_{\ell\cdot}$ is given by the group soft-thresholding operator:

$$\begin{aligned}\mathbf{A}_{\ell\cdot} = & \left(1 - \frac{\lambda}{\rho\left\|(\mathbf{Z}_4)_{\ell_1\ell_2}\left\{(\mathbf{U}_2)_{\ell_1\cdot} - (\mathbf{U}_3)_{\ell_2\cdot}\right\} - (\mathbf{E}_6)_{\ell\cdot} + (\mathbf{\Psi}_6)_{\ell\cdot}\right\|_2}\right)_+ \\ & \cdot \frac{\rho}{\lambda + \rho}\left[(\mathbf{Z}_4)_{\ell_1\ell_2}\left\{(\mathbf{U}_2)_{\ell_1\cdot} - (\mathbf{U}_3)_{\ell_2\cdot}\right\} - (\mathbf{E}_6)_{\ell\cdot} + (\mathbf{\Psi}_6)_{\ell\cdot}\right],\end{aligned} \tag{10}$$

where $(t)_+ := \max\{t, 0\}$.

Update $\mathbf{E}_1, \cdots, \mathbf{E}_9$ The slack variables are updated by solving quadratic subproblems of the same form. For each residual written as $\mathbf{B}_k - \mathbf{E}_k + \mathbf{\Psi}_k$, the subproblem is

$$\min_{\mathbf{E}_k} \quad \frac{\mu}{2}\|\mathbf{E}_k\|_F^2 + \frac{\rho}{2}\|\mathbf{B}_k - \mathbf{E}_k + \mathbf{\Psi}_k\|_F^2,$$

which gives

$$\mathbf{E}_k = \frac{\rho}{\mu + \rho}(\mathbf{B}_k + \mathbf{\Psi}_k). \tag{11}$$

In the present formulation,

$$\begin{aligned} &\mathbf{B}_1 = \frac{1}{2}(\mathbf{Z}_1 + \mathbf{Z}_2^\top) - \mathbf{Z}, && \mathbf{B}_2 = \mathbf{Z}_1 - \mathbf{Z}_2, \\ &\mathbf{B}_3 = \mathbf{Z}_3\mathbf{U}_1 - \mathbf{U}, && \mathbf{B}_4 = \mathbf{Z}_3 - \mathbf{Z}, \\ &\mathbf{B}_5 = \mathbf{U}_1 - \mathbf{U}, && \mathbf{B}_7 = \mathbf{U}_2 - \mathbf{U}, \\ &\mathbf{B}_8 = \mathbf{U}_3 - \mathbf{U}, && \mathbf{B}_9 = \mathbf{Z}_4 - \mathbf{Z}. \end{aligned}$$

For $\mathbf{E}_6$, the residual is defined row-wise as

$$(\mathbf{B}_6)_{\ell\cdot} = (\mathbf{Z}_4)_{\ell_1\ell_2}\left\{(\mathbf{U}_2)_{\ell_1\cdot} - (\mathbf{U}_3)_{\ell_2\cdot}\right\} - \mathbf{A}_{\ell\cdot}, \quad \ell = 1, \cdots, L.$$

Therefore, the update $\mathbf{E}_k = \rho(\mathbf{B}_k + \mathbf{\Psi}_k)/(\mu + \rho)$ is applied for all $k = 1, \cdots, 9$.

Update dual variables After updating all primal variables, the scaled dual variables are updated as

$$\mathbf{\Psi}_k \leftarrow \mathbf{\Psi}_k + \mathbf{R}_k, \quad k = 1, \cdots, 9, \tag{12}$$

where $\mathbf{R}_k$ denotes the corresponding primal residual defined in the next subsection.

Stopping Criteria At iteration k, define the primal residuals by

$$
\begin{aligned}
\mathbf{R}_1^k &:= \frac{1}{2}\left(\mathbf{Z}_1^k + (\mathbf{Z}_2^k)^\top\right) - \mathbf{Z}^k - \mathbf{E}_1^k, \\
\mathbf{R}_2^k &:= \mathbf{Z}_1^k - \mathbf{Z}_2^k - \mathbf{E}_2^k, \\
\mathbf{R}_3^k &:= \mathbf{Z}_3^k \mathbf{U}_1^k - \mathbf{U}^k - \mathbf{E}_3^k, \\
\mathbf{R}_4^k &:= \mathbf{Z}_3^k - \mathbf{Z}^k - \mathbf{E}_4^k, \\
\mathbf{R}_5^k &:= \mathbf{U}_1^k - \mathbf{U}^k - \mathbf{E}_5^k, \\
\mathbf{R}_{6,\ell}^k &:= (\mathbf{Z}_4^k)_{\ell_1\ell_2}\left\{(\mathbf{U}_2^k)_{\ell_1\cdot} - (\mathbf{U}_3^k)_{\ell_2\cdot}\right\} - \mathbf{A}_{\ell\cdot}^k - (\mathbf{E}_6^k)_{\ell\cdot}, \quad \forall \ell, \\
\mathbf{R}_7^k &:= \mathbf{U}_2^k - \mathbf{U}^k - \mathbf{E}_7^k, \\
\mathbf{R}_8^k &:= \mathbf{U}_3^k - \mathbf{U}^k - \mathbf{E}_8^k, \\
\mathbf{R}_9^k &:= \mathbf{Z}_4^k - \mathbf{Z}^k - \mathbf{E}_9^k.
\end{aligned}
$$

The aggregated primal residual norm is

$$
r^k := \left(\sum_{q \in \{1,2,3,4,5,7,8,9\}} \|\mathbf{R}_q^k\|_F^2 + \sum_{\ell=1}^{L} \|\mathbf{R}_{6,\ell}^k\|_2^2 \right)^{1/2}.
$$

Let

$$
m_{\mathrm{eq}} := 4n^2 + 4np + Lp,
$$

which is the total number of scalar equality constraints. Define

$$
\begin{aligned}
s_1^k := \Bigg(& \left\| \frac{1}{2}\left(\mathbf{Z}_1^k + (\mathbf{Z}_2^k)^\top\right) \right\|_F^2 + \|\mathbf{Z}_1^k - \mathbf{Z}_2^k\|_F^2 + \|\mathbf{Z}_3^k \mathbf{U}_1^k\|_F^2 + \|\mathbf{Z}_3^k\|_F^2 + \|\mathbf{U}_1^k\|_F^2 \\
& + \sum_{\ell=1}^{L} \left\| (\mathbf{Z}_4^k)_{\ell_1\ell_2}\left\{(\mathbf{U}_2^k)_{\ell_1\cdot} - (\mathbf{U}_3^k)_{\ell_2\cdot}\right\} \right\|_2^2 + \|\mathbf{U}_2^k\|_F^2 + \|\mathbf{U}_3^k\|_F^2 + \|\mathbf{Z}_4^k\|_F^2 \Bigg)^{1/2},
\end{aligned}
$$

and

$$
\begin{aligned}
s_2^k := \Bigg(& \|\mathbf{Z}^k + \mathbf{E}_1^k\|_F^2 + \|\mathbf{E}_2^k\|_F^2 + \|\mathbf{U}^k + \mathbf{E}_3^k\|_F^2 + \|\mathbf{Z}^k + \mathbf{E}_4^k\|_F^2 + \|\mathbf{U}^k + \mathbf{E}_5^k\|_F^2 \\
& + \sum_{\ell=1}^{L} \|\mathbf{A}_{\ell\cdot}^k + (\mathbf{E}_6^k)_{\ell\cdot}\|_2^2 + \|\mathbf{U}^k + \mathbf{E}_7^k\|_F^2 + \|\mathbf{U}^k + \mathbf{E}_8^k\|_F^2 + \|\mathbf{Z}^k + \mathbf{E}_9^k\|_F^2 \Bigg)^{1/2}.
\end{aligned}
$$

The primal tolerance is set to

$$
\epsilon_{\mathrm{pri}}^k := \epsilon_{\mathrm{abs}} \sqrt{m_{\mathrm{eq}}} + \epsilon_{\mathrm{rel}} \max\{s_1^k, s_2^k\}.
$$

Since the constraints include multiaffine terms, we additionally monitor the successive change of the primal variables rather than using the classical dual residual for standard ADMM. Define

$$
\begin{aligned}
\Delta^k := \Bigg(& \|\mathbf{Z}^k - \mathbf{Z}^{k-1}\|_F^2 + \sum_{j=1}^{4} \|\mathbf{Z}_j^k - \mathbf{Z}_j^{k-1}\|_F^2 \\
& + \|\mathbf{U}^k - \mathbf{U}^{k-1}\|_F^2 + \sum_{j=1}^{3} \|\mathbf{U}_j^k - \mathbf{U}_j^{k-1}\|_F^2 \\
& + \|\mathbf{A}^k - \mathbf{A}^{k-1}\|_F^2 + \sum_{j=1}^{9} \|\mathbf{E}_j^k - \mathbf{E}_j^{k-1}\|_F^2 \Bigg)^{1/2}.
\end{aligned}
$$

Let

$$
m_{\mathrm{var}} := 9n^2 + 8np + 2Lp,
$$

which is the total number of scalar primal variables monitored in Δ^k. Define

$$h^k := \left(\|\mathbf{Z}^k\|_F^2 + \sum_{j=1}^{4} \|\mathbf{Z}_j^k\|_F^2 + \|\mathbf{U}^k\|_F^2 + \sum_{j=1}^{3} \|\mathbf{U}_j^k\|_F^2 + \|\mathbf{A}^k\|_F^2 + \sum_{j=1}^{9} \|\mathbf{E}_j^k\|_F^2 \right)^{1/2}.$$

The successive-change tolerance is set to

$$\epsilon_{\text{chg}}^k := \epsilon_{\text{abs}} \sqrt{m_{\text{var}}} + \epsilon_{\text{rel}} \max\{h^k, h^{k-1}\}.$$

The algorithm is terminated when

$$r^k \le \epsilon_{\text{pri}}^k \quad \text{and} \quad \Delta^k \le \epsilon_{\text{chg}}^k.$$

The resulting optimization procedure is summarized in Algorithm 1.

Algorithm 1 Multi-affine ADMM for the proposed method

Require: Data matrix $\mathbf{X}$, edge set $\mathcal{E}$, regularization parameters λ and γ, ADMM parameter ρ, slack penalty parameter μ, tolerances ϵ_{abs} and ϵ_{rel}, and maximum iteration $K_{\max}$.

Ensure: Prototype matrix $\mathbf{U}$ and adaptive proximity matrix $\mathbf{Z}$.

```
Initialize U, U_1, U_2, U_3, Z, Z_1, Z_2, Z_3, Z_4, A.
Initialize slack variables E_1, ..., E_9 and scaled dual variables Ψ_1, ..., Ψ_9.
Set k = 0.
repeat
    Update Z using (1).
    Update Z_1 by row-wise simplex projection using (2).
    Update Z_2 by column-wise simplex projection using (3).
    Update Z_3 using (4).
    Update (Z_4)_{ℓ1 ℓ2} for each edge ℓ = (ℓ1, ℓ2) ∈ E using (5).
    Update U using (6).
    Update U_1 using (7).
    Update (U_2)_{i·} row-wise for i = 1, ..., n using (8).
    Update (U_3)_{j·} row-wise for j = 1, ..., n using (9).
    Update A_{ℓ·} for each edge ℓ ∈ E using the regularized group soft-thresholding update in (10).
    Update E_1, ..., E_9 using (11).
    Update the scaled dual variables Ψ_1, ..., Ψ_9 using (12).
    Compute r^k, Δ^k, ε^k_pri, and ε^k_chg.
    Set k ← k + 1.
until r^k ≤ ε^k_pri and Δ^k ≤ ε^k_chg.
Extract prototypes from the distinct rows of the converged matrix U.
return U and Z.
```

2 Additional Results

2.1 Data Description

Table 1: Description of the real benchmark datasets used in the experiments. The column p denotes the feature dimension after preprocessing. Numerical variables are standardized, and categorical variables are one-hot encoded when included.

Dataset	n	p	Label	# Classes
Iris	150	4	Species	3
Wine	178	13	Cultivar	3
Crabs	200	7	Sex	2
Seeds	210	7	Wheat variety	3
Palmer Penguins	333	4	Species	3
WDBC	569	30	Diagnosis	2

We use six real benchmark datasets to evaluate the prototype representation obtained by each method. The datasets vary in sample size, feature dimension, and number of classes, allowing us to assess the behavior of the learned prototypes across different data structures. Class labels are used only for evaluation and are not used in the optimization procedure. Table **??** summarizes the datasets.

2.2 Simulation Data Analysis

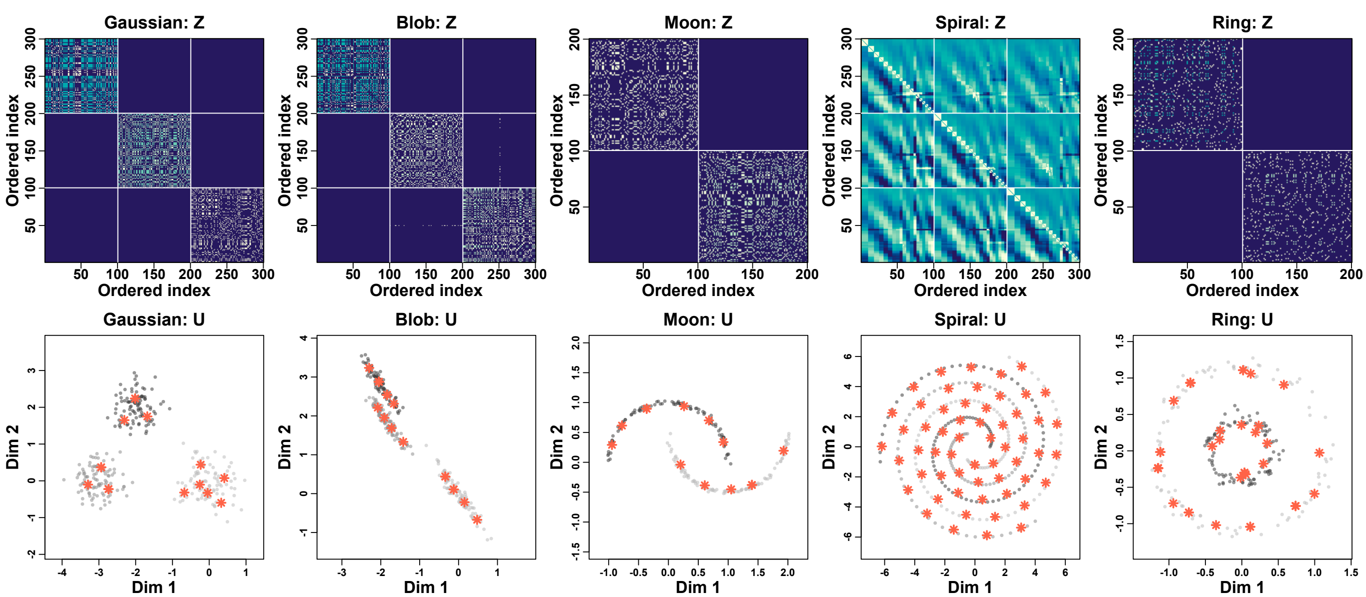

Figure 1: Additional simulation results showing the learned proximity matrix $\mathbf{Z}$ and the corresponding prototype matrix $\mathbf{U}$. The rows and columns of $\mathbf{Z}$ are ordered according to the true class labels only for visualization purposes. For the three-dimensional spiral and ring datasets, the prototype plots are visualized using only the first two coordinates.

Figure 1 provides additional simulation results for understanding the relationship between the learned proximity matrix $\mathbf{Z}$ and the prototype matrix $\mathbf{U}$. In this visualization, the rows and columns of $\mathbf{Z}$ are ordered according to the true class labels. The class labels are used only for visualization and are not used in the optimization procedure. This ordering makes it easier to inspect whether the learned proximity matrix captures the underlying group structure of the data.

The upper row shows the learned proximity matrix $\mathbf{Z}$, while the lower row shows the corresponding learned prototypes obtained from $\mathbf{U}$. In several datasets, the ordered $\mathbf{Z}$ exhibits clear block-wise patterns, indicating that observations from the same class tend to have stronger learned connections. These blocks can also be interpreted as prototype-level structures, since stronger within-block connectivity leads to fused or nearby rows in $\mathbf{U}$. The lower panels confirm this relationship: the learned prototypes summarize the major geometric structure of each dataset while remaining aligned with the data distribution.

In addition to the datasets used in the main simulation study, we include Gaussian and blob datasets to show the behavior of the proposed method under simpler cluster structures. For Gaussian and blob data, $\mathbf{Z}$ forms relatively clear block-diagonal patterns, and the prototypes in $\mathbf{U}$ are placed around the main cluster regions. For nonlinear datasets such as moon, spiral, and ring, the learned prototypes are distributed along the intrinsic geometric structure of the data rather than collapsing to simple Euclidean centers. These results illustrate that the learned proximity matrix $\mathbf{Z}$ and the prototype matrix $\mathbf{U}$ provide complementary views of the same representation-learning process: $\mathbf{Z}$ captures prototype-level connectivity, while $\mathbf{U}$ provides a compact representation induced by that connectivity.

2.3 Real Data Analysis

Figures 2 and 3 report additional real-data results obtained under different k-nearest-neighbor graph settings. These results are intended to examine whether the empirical trends discussed in the main text are sensitive to the choice of the neighborhood size used to construct the graph for convex clustering.

Overall, the additional results show a similar tendency to the main experiments. Changing the value of k affects the DR values of both convex clustering and the proposed method, since the neighborhood size directly changes the local connectivity structure used in graph-based prototype learning. Nevertheless, the two methods remain broadly comparable across the tested settings, and the relative interpretation does not substantially change. In datasets with relatively simple and well-structured geometry, convex clustering can perform competitively when the pre-specified neighborhood graph is reasonably aligned with the underlying data structure. This observation is consistent with the main results, where fixed proximity weights are sufficient for some datasets.

It is worth noting that the proposed method still shows strong performance on WDBC, which is relatively high-dimensional and contains heterogeneous diagnostic features. This result supports the interpretation that adaptive proximity learning is particularly useful when a manually specified local graph may not fully capture the latent discriminative structure of the data. Although the DR values vary with the neighborhood parameter, the proposed method remains effective in this more complex setting by jointly learning the prototypes and the proximity structure. Therefore, these additional results further support the main conclusion that the proposed method provides a flexible prototype representation, especially when the appropriate proximity structure is difficult to specify in advance.

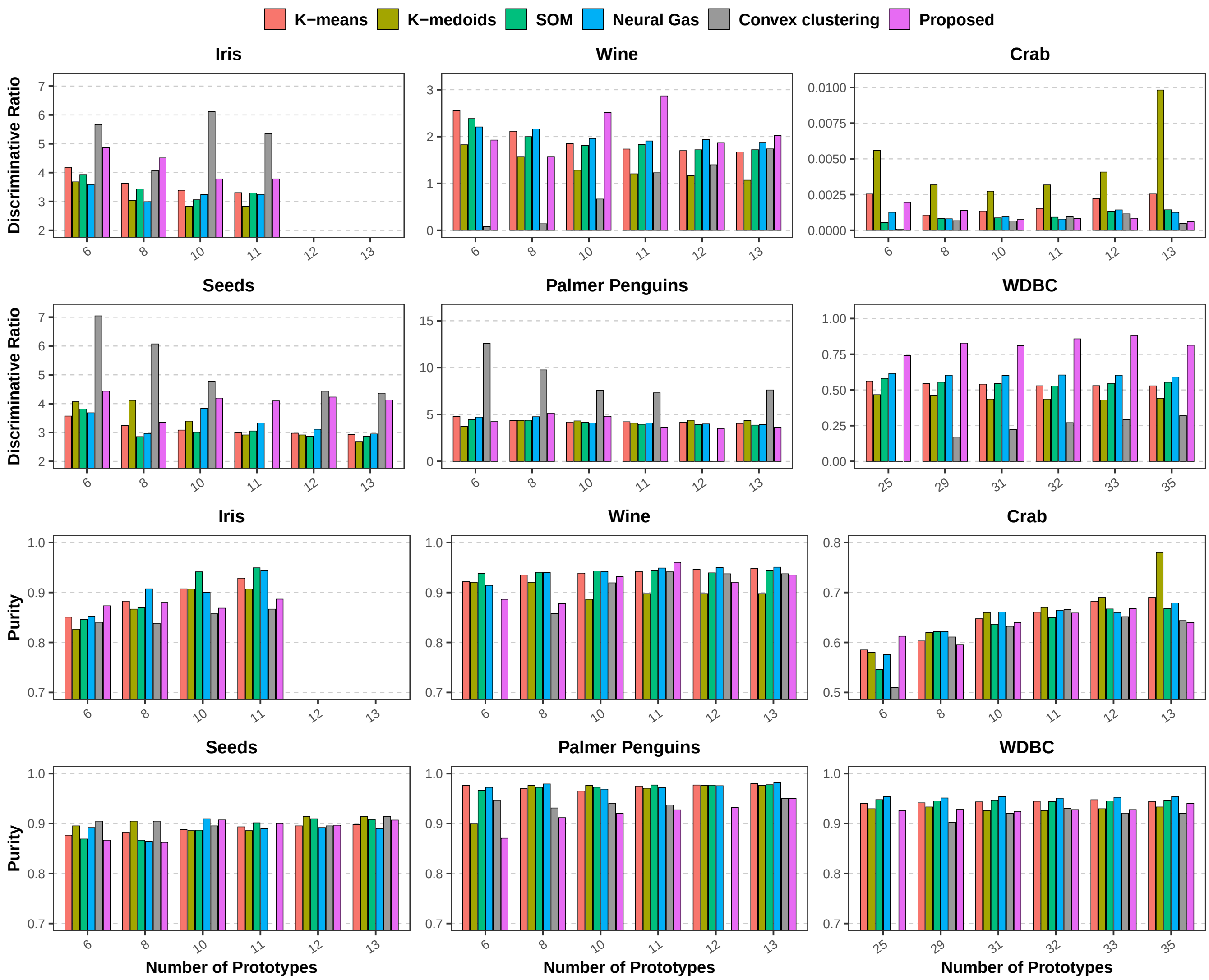

Figure 2: Prototype representation comparison on real benchmark datasets under the k-nearest-neighbor graph setting with $k = 5$. Dataset names are shown in the subplot titles. The top two rows report the discriminative ratio (DR), and the bottom two rows report purity at selected prototype counts. Bars denote averages over repeated runs, except for deterministic methods. Missing bars indicate configurations in which convex clustering or the proposed method does not produce exactly the selected number of prototypes.

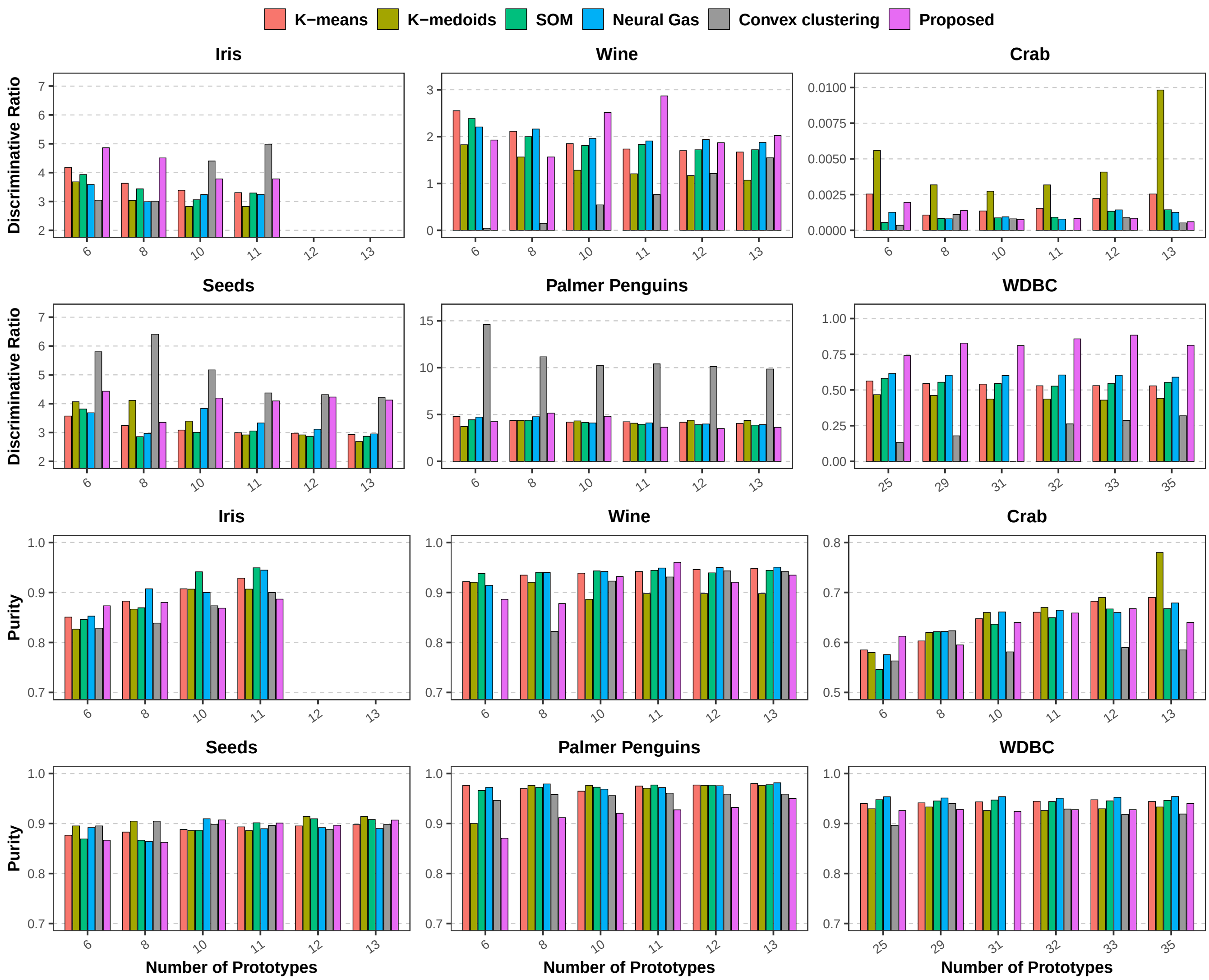

Figure 3: Prototype representation comparison on real benchmark datasets under the k-nearest-neighbor graph setting with $k = 10$. Dataset names are shown in the subplot titles. The top two rows report the discriminative ratio (DR), and the bottom two rows report purity at selected prototype counts. Bars denote averages over repeated runs, except for deterministic methods. Missing bars indicate configurations in which convex clustering or the proposed method does not produce exactly the selected number of prototypes.

References

Wenbo Gao, Donald Goldfarb, and Frank E. Curtis. 2020. ADMM for multiaffine constrained optimization. *Optimization Methods and Software* 35, 2 (2020), 257–303. doi:10.1080/10556788.2019.1683553